5G PUSCH Channel Estimation and Decoding Subject to High-Power Pulse Radar Interference

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Abstract:
The continued high demand for additional spectrum has compelled governments to consider opening up more of its spectrum to the commercial sector. Consequently, this has also put pressure on government users of the spectrum to be more efficient even as their own need for high data rates grows. As a result, the US government is contemplating using the 3.1 GHz – 3.45 GHz band for federal 5/6G networks, but also opening the band up to commercial networks. Such commercial 5/6G systems will need to coexist with multiple non-communications systems including high-power pulse radars which currently occupy this spectrum. In this work, we present a method for dealing with such high-power radar pulse interference within the data carrying 5G physical uplink shared channel. Through appropriate blanking of corrupted soft-detector outputs, acquiring a radar free noise power estimate, and deemphasizing of corrupted reference signal values in the channel estimation, it is shown that decoding performance in the presence of high-power pulse radar interference is largely restored.

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SECTION I. Introduction

The commercial sector's demand for spectrum has compelled governments to consider making more spectrum available for commercial use [1]– [3]. The desire for this spectrum is evident from the billions of dollars raised by governments for auctioned spectrum, for instance the A WS auctions in the United States. In addition to fully licensed spectrum, other shared-use spectrum has also been made available such as CBRS [4] and extension of several unlicensed bands. Governments are also interested in employing 5G private networks for their own use and for different use cases. The 3.1 GHz – 3.45 GHz band in particularly is being considered for in band sharing between 5/6G systems and non-communications incumbent government users. Current occupants of this band include aeronautic radionavigation, shipborne air defense radars, and air base, as well as airborne search and surveillance radars [5], [6], the latter of which are generally so-called pulse radars. Such pulse radars transmit a *short* frequency modulated pulse. The echoes of which are analyzed to determine the range and velocity of a particular target. Generally, these radars employ high transmission power and high gain antennas enabling operation over very large distances. As such, pulse radars are extensively being used for weather surveillance (both ground-based and airborne), air traffic control, as well as for a variety of military applications [5].

As the 5G standard was not made with radar coexistence in mind, modifications of the 5G receiver are expected to ensure satisfactory performance. It may be argued that joint design would provide the best theoretical performance for both the communication and the radar system [7]. However, even if the spectrum could wait for such a joint re-redesign, the systems' distinct requirements for waveform, power, range, and direction of operation make coexistence, rather than joint design, a more practical solution considering the cost and time associated with developing a new joint system.

Various methods have been proposed to enable coexistence between a communication system and radar, such as beamforming [8], opportunistic spectrum access [9] and power control [10] aspiring to reduce the mutual interference between the systems. The approach in this paper aims not to compete with other methods, but to provide additional protection when used in combination with these other techniques, effectively extending the range of coexistence. The focus lies on the 5G physical uplink shared channel (PUSCH) which carries data as well as the demodulation reference signal (DMRS) used to obtain both channel and noise power estimates. Hence, depending on whether the radar interference coincides with DMRS-carrying resource elements (RE) or not, the interference impact will be different. As such, the suggested method deals with both these cases by appropriately blanking corrupted log likelihood ratio (LLR) values from the output of the soft demodulator, generating an interference free noise power estimate through exclusion of corrupted DMRS, and obtaining a suitable channel estimate by deemphasizing the impact of corrupted DMRS-carrying REs.

SECTION II.
System Description

The considered setup consists of a single 5G basestation, referred to as gNB, a UE and a single pulse radar interferer, which is depicted as an airborne radar in Fig. 1, but could just as well be ground-based. In this scenario, the pulse radar uses highly directional transmission and the gNB uses receiver beamforming to increase the uplink signal to noise ratio (SNR). Despite the radars relatively suppressed sidelobes in comparison to main beam, it should be noted that transmission power can be as high as 1 MW with antenna gains of 40 dBi. As such, even when the location of the gNB does not directly coincide with the radar's main beam, substantial radar interference may be expected from the radar's sidelobes and in the adjacent carrier. It is assumed that the UE and gNB have established synchronization, and that the gNB does not have any prior knowledge of REs affected by the interfering radar but obtains this information during processing of the current slot from either its own estimates using the received signal, or from external sensors. In principle, the following techniques can also be used in the UE if the UE has knowledge of the interferer.

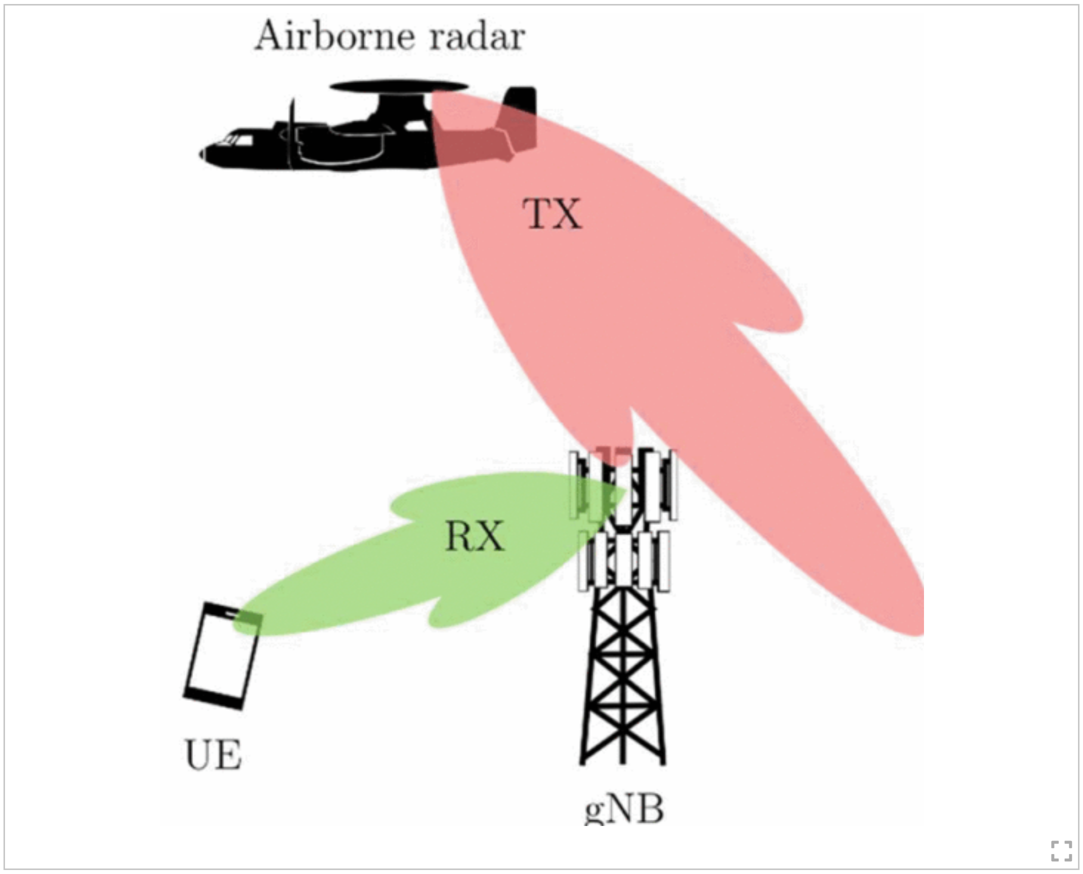

Fig. 1
Depiction of uplink reception being interfered by airborne pulse radar interference.

5G resource allocation can be represented in a time-frequency resource grid as depicted in Fig. 2, where each RE can carry one modulated symbol. Within PUSCH, there are two types of symbols, data and DMRS, where the specific DMRS mapping is determined by higher layers. Consequently, the radar interference can impact the reception in two different ways. First, if the radar interference coincides with DMRS symbols (red radar pulse in Fig. 2), both the channel and the noise estimate will be affected by the interference. As those channel estimates may be used for the entire slot, performance will be degraded until the next estimate is acquired. This is similar for the estimated noise power, which may be appropriated for the symbol with the interference, but not for other interference-free symbols. Secondly, if the DMRS is not corrupted (green radar pulse in Fig. 2), pulsed interference can still result in inaccurate combining ratio estimation for decoding schemes such as LDPC, effectively corrupting an entire codeword, that spans multiple OFDM symbols [11]. Though only PUSCH is considered here, the Physical Downlink Shared Channel (PDSCH) has a similar frame structure and would be affected in a similar way by pulse radar interference.

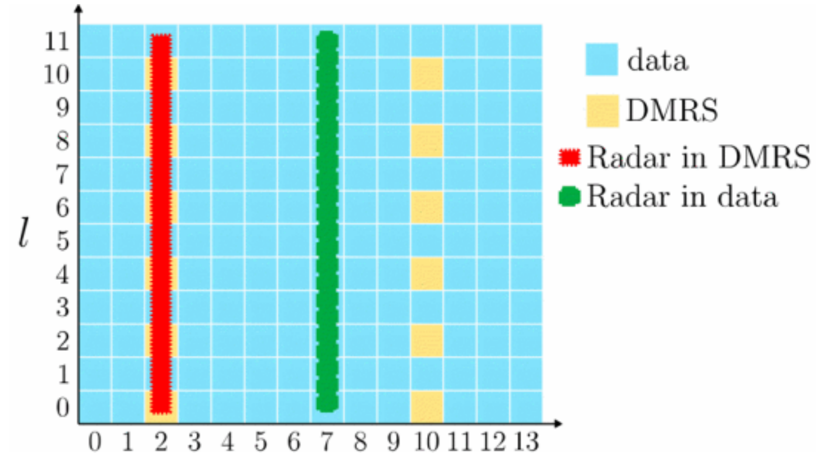

Fig. 2:
Resource element mapping for PUSCH in a single resource block, showing DMRS RES (yellow) and data res (blue). Also included is a single radar pulse either in DMRS (red) or in data (green).

A. Radar Interference Not in DMRS

When the radar pulse does not coincide with DMRS-carrying REs, neither the obtained channel estimate nor the noise power estimate needs additional processing as these are radar interference free. However, the radar interference may still result in substantial performance loss as corrupted log likelihood ratio (LLR) values fed into the LDPC decoder can result in inaccurate combining ratio estimation, affecting the decoding of entire codewords [11]. Assuming the OFDM symbols to be QPSK modulated, and the noise to be additive white Gaussian noise (AWGN), the LLR output can be written as

$$\lambda = \log \frac{P(c=0|y)}{P(c=1|y)} = \frac{2A}{\sigma^2} y,$$

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where y is the received symbol, A is the amplitude of the transmitted symbol, and σ^2 corresponds to the noise variance. The LDPC decoder iteratively refines λ before taking the decision concerning the value of a particular bit, and may, for algorithms based on the min^* function, rely on an accurate combing ratio $2A/\sigma^2$ to correctly decode bits.

In [11] this issue was handled by using the so-called *offset min sum* algorithm. Here, the combining issue is here alleviated by zeroing out the LLR output value $r_{\mathrm{LLR}}[k]$, among the K LLR outputs if it is deemed corrupted. This solution can be directly implemented with already existing LDPC decoders but falls short of the performance of the *offset min sum*. As the LLR should reflect the certainty of a particular bit-value, the LLR obtained from corrupted REs should be very low as their associated uncertainty is significant. Thus, by setting corrupted LLR values to zero, it informs the decoder that the associated bit is as likely to be a 0 as a 1. Consequently, the same corrupted value will have little impact on decoding of other bits, which aligns with the desired behavior.

However, there is a challenge associate with selecting an appropriate threshold for the decision on whether to zero out LLR values or not. The discrepancy arises when the radar interference is detectable, but not significant enough to warrant zeroing out. This case is exemplified in the left plot in Fig. 3 where it is evident that the constellation points corrupted by radar are still within their respective decision boundary and would be correctly decoded if unaltered.

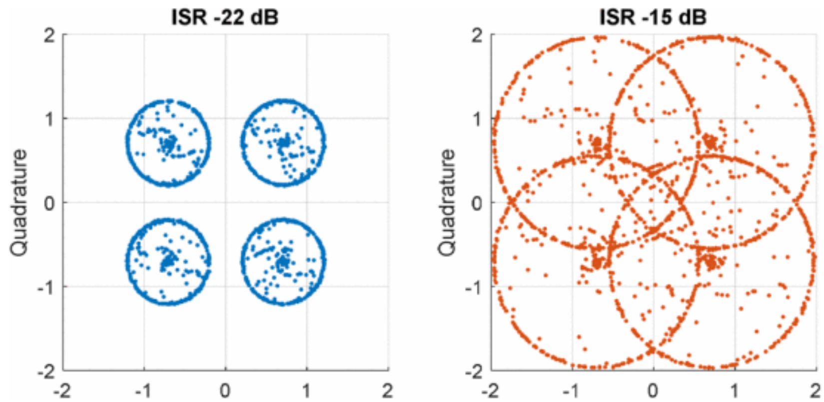

Fig. 3:
Exemplification of radar interference impact on QPSK constellation diagram for relatively low (left) and high (right) interference to signal ratio (ISR).

In contrast, in the right plot in Fig. 3, it is not as clear which symbol corresponds to which constellation point, justifying zeroing out LLR values associated with those points. As such, it possible to base the decision of zeroing out a particular LLR value on the interference power according to

$$r'_{\mathrm{LLR}}[k]=\begin{cases} r_{\mathrm{LLR}}[k], & P_{Z,k}<\alpha \\ 0, & \text{otherwise}\end{cases}$$

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where $P_{Z,k}$ is the interference power within the RE associated with LLR output value k, and α is an empirically obtained threshold value find through simulation. It should be noted that different modulations will warrant different α as the Euclidian distance between constellation points is generally smaller for higher order modulation, making these less tolerant to interference. Furthermore, it can be noted that the larger the number of corrupted REs, the more LLR outputs will be blanked. As such, a radar pulse with large bandwidth (BW), or one with high pulse reptetition frequency (PRF) has the potential to cause more LLR outputs being blanked resulting in worse system performance in terms of throughput (TP).

B. Radar Interference in DMRS

When radar interference coincides with DMRS-carrying REs, two issues arise. Firstly, the noise power estimate, used in both channel estimation and equalization, and often obtained from DMRS-carrying REs will be corrupted by radar interference and prove inappropriate for use in symbols where radar is not present. Secondly, the channel estimate obtained using corrupted DMRS symbols can be heavily degraded by the interference. Methods to avoid these effects are presented in the following.

1) Pulse Radar Free Noise Power Estimate

The assumed noise power estimation is based on extracting and calculating the power of the *late* portion of the estimated channel impulse response. Assuming any large delayed multipath components are attenuated to well below the receiver noise floor as seen in Fig. 4, the later, marked portion of the channel impulse response is appropriate for establishing a noise power estimate. The estimated impulse response in this setup is generated using the DMRS-carrying REs which implies that radar interference in these REs will influence the noise power estimate. To counter the effect of interference, it is suggested that radar corrupted DMRS symbols are simply excluded from the estimation process. This approach is adequate as long as sufficiently many radar-free DMRS-carrying REs are available, as having fewer non-corrupted REs will result in a less accurate noise power estimate.

Fig. 4:
Notional illustration of channel impulse response where the latter channel components are assumed attenuated well below the receiver noise floor.

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2) Channel Estimation in Pulse Radar Interference

The channel estimation matrix $\hat{\boldsymbol{H}}$ consists of estimated channel coefficients at each RE, where the row index in $\hat{\boldsymbol{H}}$ corresponds to the specific RE subcarrier index $l = 0, 1 \ldots, L-1$, and the columns reflects the RE time symbol index $n = 0, 1, \ldots, N-1$ in an arbitrary chosen resource grid of size $L \times N$. Using a 2-D to 1-D mapping $T : I^2 \rightarrow I$ [12] the estimated channel coefficients corresponding to the DMRS-carrying REs can be represented as the vector $\hat{\boldsymbol{h}}' = [\hat{h}_{\acute{0}}, \ \ldots, \ \hat{h}'_{P-1}]^T$, where P denotes the number of DMRS-carrying REs in the considered resource grid and

$$\hat{h}'_p = \frac{y'_p}{x'_p},$$

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denotes the initial least square estimate of the channel coefficients corresponding to the p: th DMRS-carrying RE. Similarly, the received $p : \mathbf{th}$ DMRS symbol is y'_p and x'_p corresponds to its transmitted reference symbol. Channel estimates for arbitrary REs other than that of DMRS can then be acquired through interpolation implemented through filtering. The interpolation can be expressed as

$$\hat{\boldsymbol{h}} = \boldsymbol{W}\hat{\boldsymbol{h}}', \tag{1}$$

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where $\hat{\boldsymbol{h}}$ is a length NL vector corresponding to the T mapping of $\hat{\boldsymbol{H}}$, and $\boldsymbol{W}$ is a $NL \times P$ matrix made up of filter coefficients. In the form expressed in (1) **E** rror ! **Reference source not found.**, an alternate interpretation of W is to see it as a weighting matrix reflecting how much each DMRS-carrying RE should influence a particular estimated channel coefficient. As such, a reasonable approach is to decrease the impact of DMRS-carrying REs that are corrupted by radar interference by altering the corresponding weights in $\boldsymbol{W}$. When based on 2-D Weiner interpolation, this can effectively be done by incorporating the radar interfere power $P'_{z,p}$, into the noise power estimate [13]. However, this assumes the interference is Gaussian which is not the case. Hence, a more optimal solution may exist. Following this, the 2-D Weiner associated weighting matrix can be generated according to

$$\boldsymbol{W} = \boldsymbol{\Theta}\boldsymbol{\Phi}^{-1} = \boldsymbol{\Theta}\left(\Phi_0 + \frac{1}{\hat{P}_u}\boldsymbol{Q}\right)^{-1}$$

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where Θ is the $NL \times P$ cross-covariance between the DMRS-carrying REs and all other REs, $\boldsymbol{\Phi_0}$ is the $P \times P$ auto-covariance matrix of the DMRS-carrying REs, $\hat{P}_y$ denote the estimated received signal power, and

$$\boldsymbol{Q} = \begin{pmatrix} \hat{\sigma}^2 + P_{z,0} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \hat{\sigma}^2 + P_{z,P-1} \end{pmatrix},$$

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is a diagonal matrix with the estimated noise power $\hat{\sigma}^2$ plus the corresponding radar interference power for the specific RE on its diagonal. Note that if the radar power is very high in a particular RE, there is a reduction of the corresponding entries in $\boldsymbol{W}$, resulting in the corrupted RE having less effect on the channel estimates.

When radar interference completely overpowers DMRS-carrying REs, the corrupted REs will effectively be excluded from the channel estimation. As such, increasing the interference power further, will have no additional impact on the channel estimate. However, for a proper channel estimate to be obtainable under such circumstances, multiple other non-corrupted DMRS symbols must be available within its vicinity. Note for instance that if the radar BW $\gg$ channel coherence BW, it becomes important for additional DMRS-carrying REs to be located in other OFDM symbols and not only on other subcarriers as interpolation along the frequency direction will be poor. Furthermore, if the channel is fast fading, DMRS far (in time) from the corrupted DMRS-carrying REs will also be inadequate, justifying a need for allocation of additional DMRS-carrying REs. This, however, will reduce the maximum possible TP as less REs can be allocated to data.

SECTION III.
Results

To assess the performance of the suggested technique, simulations were performed in a 5G link level test bench implemented in MATLAB. The simulations were run in accordance with the 3GPP conformance test outlined in [10] for the two fading channels TDL300C 1 00 and TDL 1 00B400, where the first number (300 and 100) denote the channel delay spread in microseconds, and the second number (100 and 400) denote the Doppler shift in Hz. The simulations were performed in the presence of a FM pulse radar, configured to transmit one radar pulse per slot with interference to signal ratio (ISR) 0 dB, and use a BW of 50 MHz. These parameters do not correspond to any particular radar system but are within the range of typical values that might be expected for the 3.5 GHz band [6]. The DMRS mapping used was A1, with one additional DMRS time occurrence as shown on the left in Fig. 5. The performance was evaluated in terms of TP as a function of SNR for a given radar pulse location within the slot. For comparison, the 'no interference' case and the 'no mitigation' case were also included.

Fig. 5:
Comparison between al addl single and double DMRS mapping, with radar present.

In Fig. 6, the achieved TP for modulation and coding scheme (MCS) 8 is shown for radar in DMRS and radar in data with and without the proposed mitigation. In either radar pulse location, the TP drops to zero for all SNR (orange curve) with no mitigation. The yellow triangle curve Fig. 6. corresponds to the radar pulse occurring within the first DMRS allocated OFDM symbol, as shown in the left of Fig. 5, while for the 'Radar in data' curve (purple x) the radar pulse occurs in the 6^{th} OFDM symbol. The observed degradation is between 1–2 dB when compared to the 'no interference' case. When radar coincide with the data, the TP is slightly worse as more data-carrying REs are affected by radar and their corresponding LLR outputs are blanked.

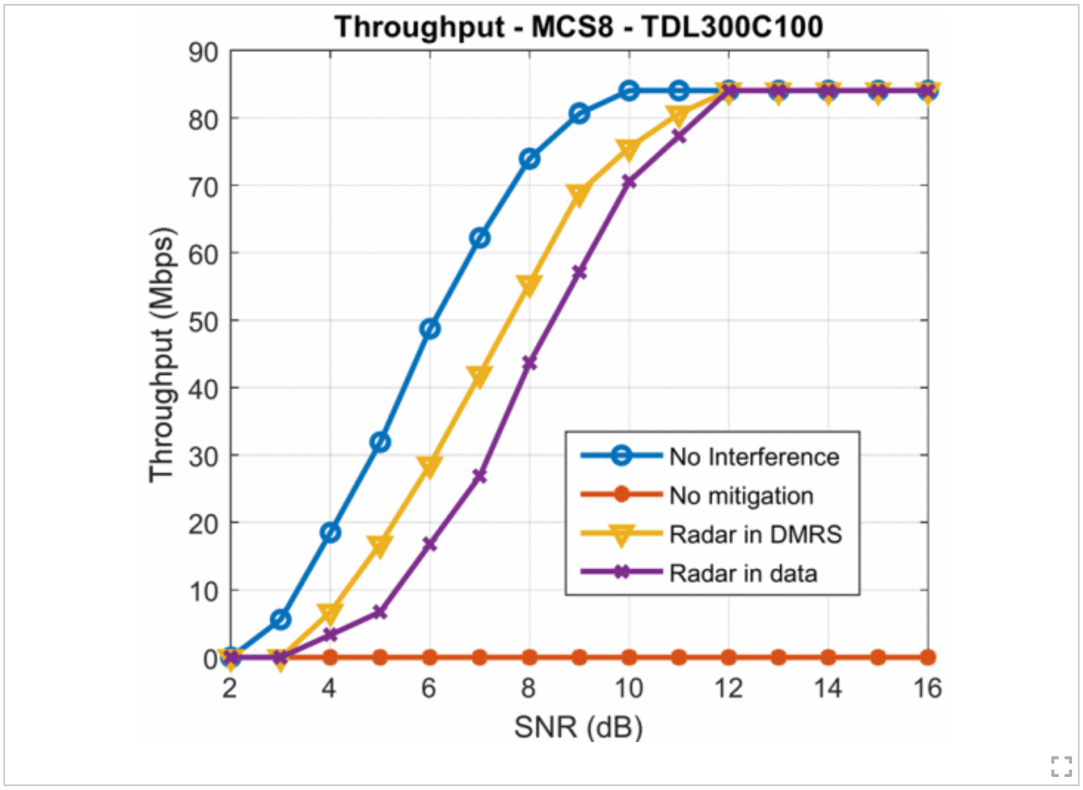

Fig. 6:
Achieved throughput performance mcs8 (qpsk), single pulse 50 mhz bandwidth fm radar for tdl300c100 channel.

Using the same radar parameters, the test was repeated for MCS14 (16-QAM) shown in Fig., where having radar in the DMRS had a similar impact. However, when the radar coincides with the data, there is a much larger performance degradation compared to MCS8. The reason for this is that the higher MCS 14 has less redundancy (higher coding rate), making it more susceptible to errors and not able to recover as well when many LLR values are blanked.

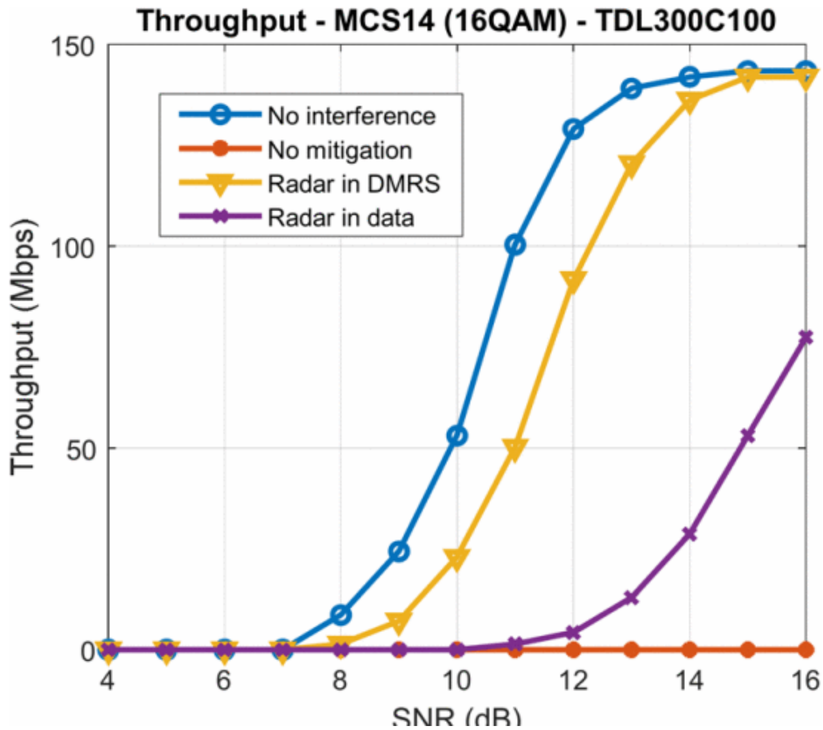

Fig. 7:
Achieved throughput performance MCS14 (16QAM), single pulse 50 mhz bandwidth fm radar for TDL300C100 channel.

The TDL 1 00B400 channel has shorter coherence time and results are shown in Fig. 7. With the shorter coherence time, it is important to have interference free DMRS in both the early and late parts of the slot as shown by the complete TP loss of the 'Radar in DMRS single' curve. The DMRS mapping on the left-hand side of Fig. 5 does not suffice for such a time varying channel when radar coincide with the first DMRS allocated OFDM symbol. This is because the channel changes over the slot duration to the degree that channel estimation for the REs far from the interference free DMRS-carrying REs are not adequate. In contrast, the 'Radar in data' curve does not suffer the same performance loss since a suitable channel estimate can be obtained for that case. To counter the effect of the small coherence time channel and radar interference in DMRS, the DMRS mapping of type *double* (right mapping in Fig. 5) was investigated. This double-symbol is available in 50 in support of enabling multiple antenna ports, and not a normal configuration for single-layer transmission. Nevertheless, with the ability to utilize such a mapping for pulse interference mitigation, it can, in this case be ensured that there are always radar free DMRS symbols available, closely located in time. Hence, assuming such a mapping, the system is able to recover most of the TP (green square curve). However, increasing the number of DMRS-carrying REs, means reducing the possible number of data-carrying REs which is shown in Fig. 7, as the maximum TP is lowered from 84 Mbps to 78 Mbps.

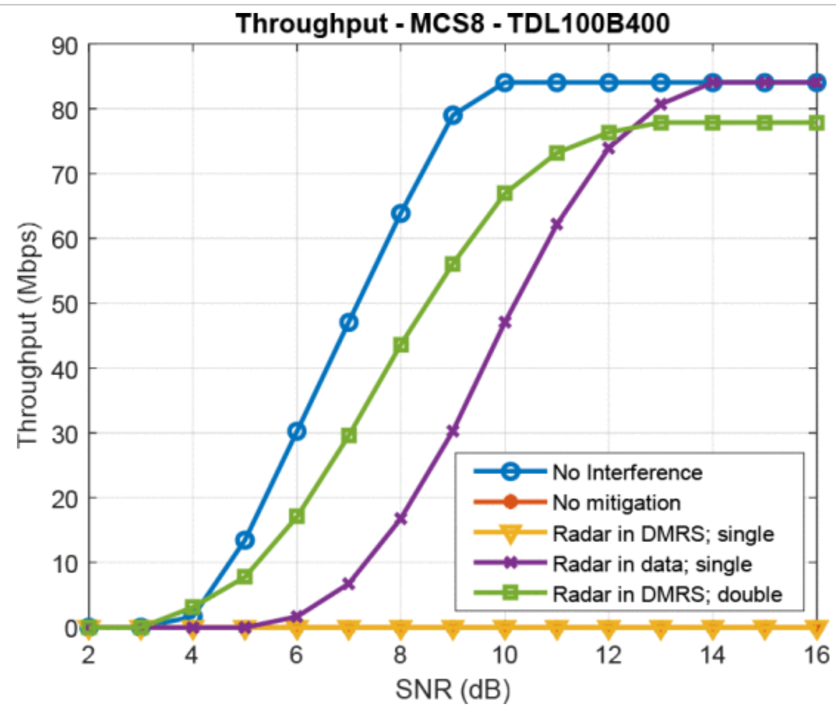

Fig. 7:
Achieved throughput performance MCS8 (QPSK), single pulse 50 mhz bandwidth fm radar for TDL100B400 channel.

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SECTION IV.
Conclusion

Motivated by the potential use of the 3.1 GHz to 3.45 GHz band for commercial cellular services on a coexistence basis, we have studied and proposed methods to aid in coexistence of the 5G PUSCH with high-power pulsed radars. The approach involved blanking corrupted LLR output values from the soft demodulator, acquiring an interference free noise power estimate by excluding corrupted DMRS, and including interference scaling in the MMSE channel estimator. Using these countermeasures, reception of otherwise lost transmissions was possible.